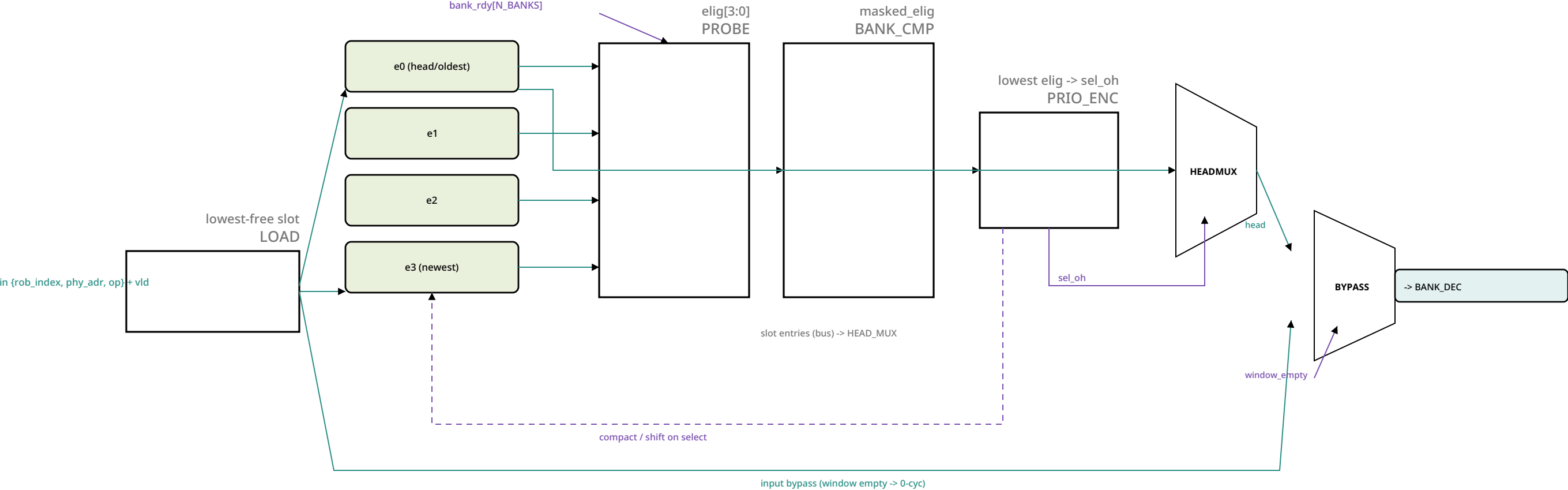


LOOKAHEAD - compacting queue + parallel prio_enc + bypass mux



Depth 4 = reorder window, NOT latency. New entry -> lowest-free slot; all slots probed vs bank_rdy in parallel; bank_cmp holds per-bank order; prio_enc picks lowest eligible. BYPASS forwards input->out when window empty (0-cyc); else 1-cyc.